NOT MEASUREMENT SENSITIVE

MIL-C-70987 (AR)
01 December 1994

MILITARY SPECIFICATION

CARTRIDGE, 25MM, TARGET PRACTICE
DISCARDING SABOT WITH TRACER (TPDS-T), M910, REFERENCE

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for one type of percussion primed reference cartridge, designated as Cartridge, 25mm, TPDS-T, M910 to be used for calibrating acceptance testing equipment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1305

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SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-50603	-	Primer, Percussion, M115

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-651	-	Visual Inspection Standards for 20mm Ammunition and Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9391399	-	Interface Control Drawing, Max. Profile and Alignment, Cartridge, M791
12013671	-	Shipping Container, Linked, 25mm Ammunition (50 rounds)
12013697	-	Interface Control Drawing, 25mm Weapon/Ammunition
12524001	-	Barrel Assembly
12524507	-	Barrel Assembly
12556500	-	Cartridge, 25mm, TPDS-T, M910

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PUBLICATIONS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

AS 12013566 - 25 Millimeter Ammunition Ballistic Test Methods

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Material, design and construction. The cartridge material, design and construction shall be in accordance with MIL-A-2550 and drawing (dwg.) 12556500. The primer used in the cartridge shall be within the family of the M115 Percussion Primer conforming to MIL-P-50603.

3.1.1 Manufacturing process. All parts and assemblies shall be manufactured by a conventional manufacturing process. Any deviations from conventional manufacturing processes shall be approved by the contracting officer. Any change in source of supply shall be approved by the contracting officer.

3.2 25mm weapon/ammunition interface. Performance and acceptance as required in this specification apply only when tests are conducted in weapons which conform to the requirements of dwg. 12013697. Autogun test barrels shall be in accordance with dwg. 12524001 or 12524507. Testing from a firm fixed mount is required.

3.3 Projectile torque. The projectile shall be capable of withstanding a minimum torque of 30 inch-pounds without rotational movement with respect to the cartridge case (see 4.4.2).

3.4 Projectile extraction. The tensile force necessary to extract the projectile from the cartridge case shall be as specified on dwg. 12556500 (see 4.4.3).

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3.5 Propellant contamination. The propellant shall not contain foreign matter in excess of the requirements listed on the respective propellant drawing (see 4.4.4).

3.6 Propellant weight. The loaded cartridge propellant weight shall be such that the requirements of 3.7, 3.8 and 3.9 are satisfied.

3.6.1 Charge weight. The charge weight of the cartridges shall be maintained within ± 0.1 gram of the established charge weight (see 4.5).

3.7 Chamber pressure. The average plus 3.0 standard deviations of peak chamber pressure shall not exceed 454 megapascals (MPa) when the cartridge is conditioned and fired at 21° ± 3°C from a Mann Barrel. The average plus 3.0 standard deviations shall not exceed 496 MPa, with no individual measurement exceeding 496 MPa, when the cartridge is conditioned at any other temperature in the range of -54°C to +71°C. Pressure shall be measured with a piezoelectric type pressure transducer (see 4.4.5).

3.8 Action time. The action time of the cartridge shall not exceed 6.0 milliseconds when conditioned at any temperature from -54°C to +71°C and fired from a Mann barrel. Action time is defined as the time period between the initial contact of the weapon firing pin against the primer and the exit of the projectile from the muzzle (see 4.4.5).

3.9 Velocity. The average muzzle velocity shall not vary more than ± 20 meters per second (mps) from that noted on the contractor's top level drawing, listed on dwg. 12556500, when the cartridge is conditioned at 21° ± 3°C and fired in a 25mm Mann Barrel. In addition, the cartridge standard deviation shall not exceed 10 mps (see 4.4.5).

3.10 Waterproofness. There shall be no evidence of water entry into the propellant of any cartridge (see 4.4.6).

3.11 Dispersion - Mann barrel. The standard deviation(s) of projectile impacts in both the horizontal and the vertical directions of the M910 candidate reference cartridge, conditioned at +21°C, shall not be greater than 0.38 mils when fired single shot from an unported Mann Barrel on each of three consecutive days. In addition, homogeneity of variance for the three targets per day shall be proven for each of the three days (see 4.4.7).

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3.12 Projectile trace. The projectile shall provide a visible trace in both daylight and night conditions at ranges from a minimum of 100 to at least 2000 meters from the muzzle (see 4.4.8).

3.13 Function, casualty and metal parts security. The cartridge shall function without casualty or hazards when conditioned and fired from the 25mm M242 Weapon at 200 spm. The cartridges shall be conditioned to ambient temperature. The cartridge shall exhibit metal parts integrity in the weapon feed mechanisms (chuting, feeding and chambering). Components designed for separation such as the sabot, protective cap and base are excluded (see 4.4.9).

3.14 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil, corrosion products and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All markings and stampings shall be neat and legibly defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 and MIL-A-2550 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the requirements and test method paragraphs in the specification.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. The inspection lot shall comply with the lot formation provisions of MIL-A-48078. In addition, each cartridge lot offered for acceptance shall contain:

a. Primers from one lot interfix from one manufacturer.

b. Metal parts from one lot interfix from one manufacturer.

c. Boosters from one lot interfix from one manufacturer.

d. Propellant from one lot from one manufacturer.

e. Tracer mixture from one lot from one manufacturer.

f. Projectile, sabot and protective cap molding from one lot interfix from one manufacturer.

g. Cartridge cases from one lot interfix from one manufacturer.

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4.3.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II. Attributes sampling inspection.

	Inspection Levels	
Lot Size	III	IV
2 to 8	*	*
9 to 15	*	*
16 to 25	*	*
26 to 50	*	32
51 to 90	50	32
91 to 150	50	32
151 to 280	50	32
281 to 500	50	48
501 to 1200	75	73
1201 to 3200	116	73
3201 to 10000	116	86
10001 to 35000	135	108
35001 to 150000	170	123
150001 to 500000	200	156
500001 and over	244	189

NOTE: Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70987 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Cartridge, 25mm, Target Practice Discarding Sabot with Tracer (TPDS-T), M910, Reference	1 OF 2	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined.			
SPECIAL	1/			
1	Split or perforated case	100%	3.1	Visual
2	Crimp missing (case to projectile)	100%	3.1	Visual
3	Low propellant weight 2/	100%	3.6	SMTE, Probe
4	Cracked sabot segment or other plastic parts	100%	3.1	Visual
5	Profile and alignment	100%	3.1	Dwg. 9391399 Para 4.4.1
MAJOR				
101	Overall length, maximum	Level III	3.1	SMTE
102	Improper depth of primer seating	Level III	3.1	SMTE
103	Cocked, loose or inverted primer	Level III	3.14	Visual
104	Waterproofing missing around primer	Level III	3.14	Visual
105	Improper head configuration	Level III	3.14	Visual
106	Evidence of corrosion	Level III	3.14	Visual
107	Cartridge case failures (i.e., case draw, case dent, fold, bulge, wrinkle, or buckle)	Level III	3.14	Visual

NOTES:

1/ MIL-STD-651 shall apply in defining and evaluating visual defects, as applicable.

2/ Propellant height in cartridge prior to inserting projectile assemblies.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.2.1	Cartridge, 25mm, Target Practice Discarding Sabot with Tracer (TPDS-T), M910, Reference	SHEET 2 OF 2	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR	1/			
201	Marking incorrect, incomplete illegible or missing	Level IV	3.14	Visual
202	Presence of foreign matter	Level IV	3.14	Visual
203	Evidence of poor workmanship	Level IV	3.14	Visual

NOTES:

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4.3.3 Testing. The tests listed in Table III shall be performed on each cartridge lot conforming to 4.3 in accordance with the methods of inspection prescribed in 4.4. Unless otherwise indicated in the methods of inspection, tests shall be conducted with samples at 20° ± 10°C. Sample size and acceptance criteria for each test shall be as specified. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests. The combining of tests is permitted. The lot shall be rejected if in any firing test one or more projectiles remain in the gun bore. If a firing defect of Table IV occurs which would prevent the obtaining of a reliable result for the characteristics being tested, a replacement shot shall be fired. These firing defects or casualties found, whether from a Mann barrel or M242 weapon, shall be added to any found during function, casualty and metal parts security testing (see 3.16).

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TABLE III. Tests for quality conformance inspection.

Test	No. of Sample Units	Requirement Paragraph	Method Paragraph	Acceptance Criteria
Projectile torque	20	3.3	4.4.2	[1]
Projectile extraction	20	3.4	4.4.3	[1, 2]
Propellant contamination	20	3.5	4.4.4	Acc-0/Rej-1
Chamber pressure				
-54° ± 3°C	180	3.7	4.4.5	[3]
21° ± 3°C	180	3.7	4.4.5	[3]
+71° ± 3°C	180	3.7	4.4.5	
Action time				
-54° ± 3°C	180	3.8	4.4.5	Acc-0/Rej-1
21° ± 3°C	180	3.8	4.4.5	Acc-0/Rej-1
+71° ± 3°C	180	3.8	4.4.5	Acc-0/Rej-1
Velocity				
21° ± 3°C	180	3.9	4.4.5	[4]
Waterproofness				
21° ± 3°C	20	3.10	4.4.6	Acc-0/Rej-1
Dispersion [5]				
21° ± 3°C	180	3.11	4.4.7	[6]
Projectile trace				
21° ± 3°C	35	3.12	4.4.8	[7]
Function, casualty and metal parts				
Ambient [8]	200	3.13	4.4.9	Table IV

Refer to next page for "notes."

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TABLE III. Tests for quality conformance inspection. (NOTES)

[1] Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the minimum requirement.

[2] Failure of four or more sample cartridges to comply with the maximum requirement shall be cause for rejection of the lot. If more than one, but less than four sample cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined sample fail to comply with the maximum requirement.

[3] If the sample fails to comply with the chamber pressure requirement, the lot shall be rejected.

[4] If the sample fails to comply with the velocity requirement, the lot shall be rejected.

[5] Fire from a Mann Barrel with muzzle brake at a range between 100 to 200 meters.

[6] If the dispersion of the sample cartridges exceeds the applicable requirement, the lot shall be rejected. In addition, should the variances of the three targets in one or more day's firings prove to be non-homogeneous, the lot shall be rejected.

[7] The lot shall be accepted if no failures occur in the first sample. The lot shall be rejected if two or more cartridges in the first sample fail to comply with the minimum requirements. If one cartridge of the first sample fails to comply, a second sample of 35 cartridges shall be tested. The lot shall be rejected if three or more cartridges of the combined sample fail to comply with the minimum requirements. These provisions apply to blinds and/or short burn times.

[8] The cartridge lot shall be rejected if any malfunction or firing casualty of Table IV occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number. The cartridge lot shall also be rejected for the conditions listed in 4.5.11.

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TABLE IV. Firing defects.[5]

Defect Description	Acc	Rej
Cartridge:		
Misfire	0	1
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
In bore sabot separation	0	1
Primer:		
Primer leak	8	9
Primer perforation	2	3
Loose primer	0	1
Blown primer	0	1
Case:		
Longitudinal split [1]		
H or S	10	11
G or J	3	4
K, L or M	0	1
Circumferential rupture (partial) [1]		
S, J or K	3	4
G or L	2	3
Circumferential rupture (complete) [1]	0	1
Detached metal [2]	0	1
Projectile:		
Failure of sabot to separate [3]	0	1
Metal parts separation [4]	0	1
Trace fallout	1	2

Refer to next page for "notes."

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TABLE IV. Firing defects. (NOTES)

[1] See Figure 1 for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table IV for the areas involved shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classified as a complete circumferential rupture.

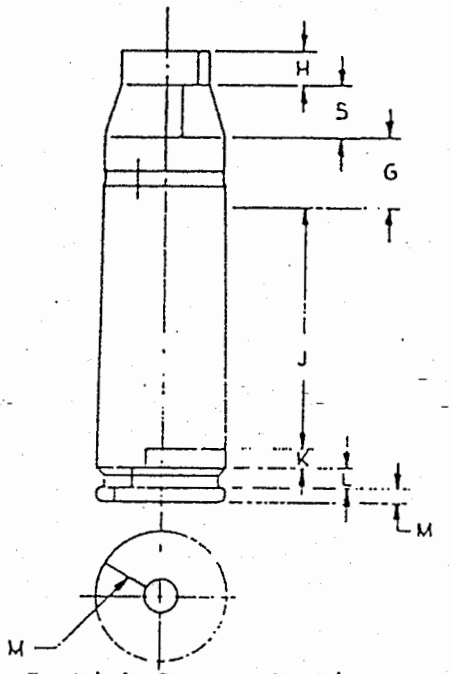
[2] Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

[3] Failure of the sabot to separate, as evidenced by penetration of the target screen, shall be classified as a defect.

[4] Break up of metal parts as evidenced by evaluation of holes in paper screen, shall be classified as a defect.

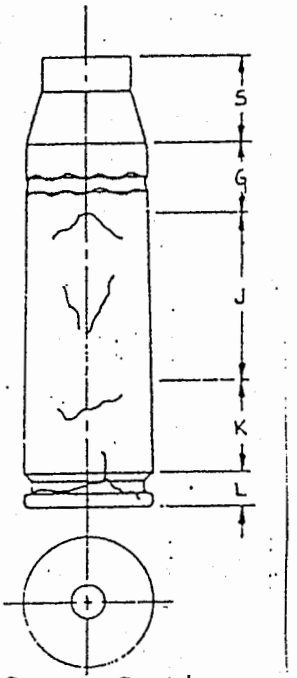
[5] Defect definitions are contained in Section 11 of AS 12013566.

SPLITS



G - In Link Groove Section
H - In Mouth or Neck
J - In Main Body Section, No Closer than 1/4 inch to Extractor
K - In Lower Body Section From Forward Edge of Extractor Groove
L - In Extractor Groove
M - In Case Head
S - In Shoulder Section

RUPTURES



G - In Link Groove Section
S - In Neck and Shoulder
J - In Upper 3/4 Body Section
K - In Lower 1/4 Body Section Above Extractor Groove
L - In Head and Extraction Groove

FIGURE 1. Classification of defects for case, cartridge, 25mm.

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4.3.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.3.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.3.4.1 Packaging inspection. The sampling and inspection of the packaging, packing and marking for conformance with 5.1 and 5.2 shall be in accordance with Drawing 12013671.

4.3.4.2 Unlisted firing defects. Ammunition lots shall be rejected and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any testing indicates that the lot is unsuited for the purpose intended.

4.4 Methods of inspection.

4.4.1 Profile and alignment, max. Each cartridge assembly shall be inserted into the chamber gage which conforms to Drawing 9391399 to check for profile and alignment. The maximum force to insert the cartridge assembly into the gage shall be 60 pounds. Discrepant cartridges shall be removed from the lot.

4.4.2 Projectile torque. Each cartridge of the test sample shall be marked with a black ink marker. The mark, shall extend axially across the projectile assembly to the case. The torque specified shall be applied gradually. Movement of the projectile assembly with respect to the cartridge as evidenced by the misalignment of the mark shall be classified as a defect. Defects in excess of the acceptance numbers shall be cause for rejection of the lot.

4.4.3 Projectile extraction. The method of test shall be as specified in AS 12013566, Section 12. Failure to satisfy requirements shall be cause for rejection of the lot.

4.4.4 Propellant contamination. The cartridge assemblies shall be examined visually for contamination as prescribed in Section 12 of AS 12013566. Evidence of contamination in excess of that prescribed on the respective propellant drawing shall be cause for rejection of the lot.

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4.4.5 Velocity, pressure and action time. The sample cartridges shall be tested in accordance with Section 5 of AS 12013566. The cartridges shall be conditioned for a minimum of four hours at -54° ± 3°C, 21° ± 3°C, and 71° ± 3°C as applicable. Reference cartridge correction factors shall not be applied to the test cartridges. Twenty (20) test samples shall be fired from each of three (3) different Mann barrels on each of three (3) consecutive days (a minimum of twelve (12) hours is required between tests on successive days). Failure to satisfy the requirements shall be cause for rejection of the lot.

4.4.6 Waterproofness. The cartridges shall be conditioned in accordance with Section 5 of AS 12013566. The cartridges shall be disassembled and the projectiles, propellant, and cartridge case interior shall be examined under ultraviolet light for evidence of leakage. Evidence of water entry into the propellant shall be cause for rejection of the lot. Leakage paths and other effects of water entry shall be noted. Photographs shall be taken as required.

4.4.7 Dispersion. The cartridges shall be tested in accordance with AS 12013566, Section 7. Firing shall be done to determine dispersion characteristics from a firm fixed mount, unless otherwise specified. All cartridges shall be fired at an approximate 30-foot by 30-foot vertical target or other suitable target to insure that all cartridges fired shall impact the target. Any round that fails to hit the target shall be cause for rejection of the lot. The target shall be flat without sagging or wrinkles. Before testing, a sufficient number of spotting and warming cartridges shall be fired to condition the weapon and to ensure that the cartridges are likely to hit the target. Impact coordinates for each cartridge, weapon functioning and other pertinent information shall be recorded. The standard deviation of each cartridge group shall be recorded. The standard deviation of each cartridge group shall be calculated in both the horizontal and vertical directions. The computation shall be in accordance with Section 7 of AS 12013566. Discrepancies from requirements shall be cause for rejection of the lot. Methods of analysis of variance shall be as stated in paragraph 6.6.

4.4.7.1 Mann barrel test. The sample cartridges shall be conditioned for a minimum of four hours at 21° ± 3°C and tested at 100 to 200 meters. Twenty (20) test samples shall be fired from each of three (3) different unported Mann barrels on each of three (3) consecutive days (a minimum of twelve (12) hours is required between tests on successive days).

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4.4.8 Projectile trace. The cartridges shall be tested in accordance with Section 8 of AS 12013566. The cartridges shall be conditioned for a minimum of four hours at 21° ± 3°C and fired from the M242 Weapon in the single shot mode. Weapon elevation will ensure a minimum of a 4000 meter projectile free flight before ground impact. Time of trace and range of trace functioning shall be determined as measured from the muzzle of the weapon. Blinds, tracer fall-outs and short burn times in excess of the acceptance numbers are not permitted and shall be cause for rejection of the lot.

4.4.9 Function, casualty and metal parts security. The cartridges shall be tested in accordance with AS 12013566, Section 11. The M242 Weapon used shall have a barrel with greater than 4,000 rounds fired through it. A vertical witness screen shall be positioned approximately 60m to 200m normal to the line of fire such that metal part failure data may be obtained. The cartridges shall be conditioned for a minimum of four hours at 21° ± 3°C. When ambient temperature conditions are 20° ± 10°C, testing can be at range ambient conditions. All firings shall be at 200 ± 50 spm in bursts of 21 to 28 rounds. The weapon shall be cooled to ambient temperature after each temperature and range group test is completed (100 maximum cartridges fired after which cooling to ambient is required.) The witness screen shall be examined during the cooling period for fragment imprints and metal parts security failures. Evidence of yaw or other erratic flight characteristics shall be recorded and evaluated for cause. Witness screen replacement shall be provided as needed. Firing defects specified in Table IV shall also be determined during the cooling period. Any weapon stoppages obtained shall be recorded and evaluated for cause. Firing defects in excess of the quantities listed in Table IV shall be cause for rejection of the lot. The lot shall be rejected if 20 percent of the rounds fired exhibit more than 15 degrees yaw.

4.5 Uniformity of charge. Once the propellant charge has been established, a loading/charge machine shall be selected which has shown the best uniformity in charge weight during a production run. Only one loading machine shall be used in the loading of the reference cartridges. Every thirty (30) minutes during the machine loading of the lot of reference cartridges, five (5) sample cartridges shall be taken from the machine and the propellant weight measured to the nearest ± 0.1 gram. Should the charge weight of any cartridge not meet the requirement that thirty minute production shall be removed from the lot.

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5. PACKAGING

5.1 Packing level A. The cartridge shall be linked into a belt and packed in accordance with instructions from the procuring agency. Linked ammunition shall be packed in accordance with Drawing 12013671 unless provided with other instruction by the procuring agency.

5.1.1 Packing level B. Same as 5.1.

5.1.2 Packing level C. Same as 5.1.

5.2 Marking. Marking on the packed container shall be in accordance with the Drawing 12013671 as appropriate.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridges covered by this specification are intended for use as reference cartridges for calibration purposes in 25mm cartridge acceptance testing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-C, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Submission of alternate quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality assurance conformance provisions shall be submitted by the contractor to the contracting activity.

6.6 Submission of homogeneity tests. Methods used to ascertain the homogeneity of variances for the dispersion tests shall be approved prior to use. Methods should be sent to ARDEC, AMSTA-AR-QAC-C, Picatinny Arsenal, NJ 07806-5000.

6.7 Subject term (key word) listing.

Small Arms
Training
Testing Cartridges

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-0107)

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